

Engine prevention plan

Validation of the Došlap engine on the owner's two running injuries, and the phased plan (A → B → C) to make it preventive rather than descriptive. Engine v0.7.3 → v0.8.

Why	ITBS (right) on 25 Apr 2026 during a long run; left groin / adductor pain from mid-September that rose after a half marathon on 19 Sep and now limits movement.
How validated	The current engine (v0.7.3, capacity model v3 + Trénink guidance) replayed day by day on the owner's real data (read-only copy of production), each morning seeing only what was known then — activities before that day, that morning's watch data, check-ins and ratings submitted so far.
Verdict	It saw the load build-up both times but (1) labelled it „nízké riziko“, (2) had no notion of pain that limits movement, of races or of recovery after them, (3) let an untested maximal effort raise capacity, and (4) was not credible in the first weeks (rest every day). The injuries themselves never reached the engine.
Status	Plan approved by the owner on 26 Sep 2026: phases A, B and C, then the open app feedback, one deploy. Implemented in engine v0.8.0 — see sections 6 and 7 for what was built and the replay afterwards.

Caveat: one runner, two events. The replay shows where the engine is blind; it is not a statistical validation, and the changes follow the literature and plain safety logic — they are not tuned to reproduce these two events.

1. What the engine said — ITBS, right leg (25 Apr 2026)

Data started on 22 Mar, so the engine had ~5 weeks of history. The 21.1 km run was 29 % longer than the longest run of the previous 30 days (16.4 km, 19 Apr, hilly: ↑394 m ↓395 m).

Morning of	State · risk	Load	Trénink would say	What happened
14–19 Apr	Stabilní · nízké	14–23	volno every day (targets 0–13 km)	3 runs incl. 16.4 km hilly
20 Apr	Přetížení · nízké	63	volno	—
25 Apr	Přetížení · nízké	64	volno (no running)	21.1 km long run → ITBS
26–30 Apr	Přetížení · nízké	28–29	volno	30 Apr: 16 km hilly (↓356 m) — injury unknown to the app

Gaps: the risk label contradicted the state; in the first weeks the guidance said „volno“ for 12 days in a row (capacity from too little history, the cycle placed as a recovery week, HR zones estimated without a birth year) — advice that gets ignored; the injury was never recorded, so the hilly run 5 days later passed unnoticed; „prior injury: ITB“ has no date and is therefore ignored by the engine (no injury-history margin).

2. What the engine said — left groin / adductor (September 2026)

Week of 7 Sep: 49 km (highest ever) incl. 21.1 km on 12 Sep at 4:53/km, HR 165 (long runs are usually 6–7 min/km). 19 Sep: Ústí half marathon at 4:59/km, HR 173, RPE 10 — two maximal 21 km efforts in 8 days (96 and 106 min in Z4+).

Morning of	Readiness	State · risk	Trénink would say	What happened
9–11 Sep	20–27 %	Stabilní · nízké	regenerace / volno	easy days
12 Sep	49 %	Stabilní · nízké	volno	21.1 km at race pace
13–18 Sep	20–63 %	Přetížení · nízké	volno / regenerace	pain felt, not logged
19 Sep	66 %	Stabilní · nízké	kvalitní, up to 20 km	half marathon race (RPE 10)
21 Sep	54 %	Stabilní · nízké	lehký běh up to 19 km	pain at 3 sites, soreness 8
22 Sep	58 %	Stabilní · nízké	lehký běh up to 14 km	—
26 Sep	—	Stabilní · sledovat	volno (recurring pain)	pain limits movement

Gaps: race day was green-lit because the 12 Sep maximal effort had already raised the intensity capacity (no pain was logged, and no report counts as „tolerated“) and had left the 7-day window; after the race (RPE 10, soreness 8, three painful sites) there was no recovery period — a light run up to 19 km was offered; pain was

logged as 2–3/10 while it limited movement, so every numeric threshold (3, 5) stayed silent until the third day of recurrence; the risk label again stayed „nízké“ during a week of „Přetížení“.

3. The plan

Phase A — symptoms and function first

#	Change	Rule	Basis
A1	Function in the daily check-in	Three yes/no questions: pain limits ordinary movement / walking; a run was shortened or changed because of pain; limping. Any „yes“ at a running-relevant site = injury-level: no running, pain-free cross-training only, physio recommended (48 h when movement is limited).	OSTRC overuse questionnaire (Clarsen 2020): participation, volume and performance capture injuries a pain score misses.
A2	Acute overload after a run	Pain at ≥ 2 sites, or soreness ≥ 7 , or RPE 9–10 with any pain → next day no running, then easy only; no waiting for 3 days of recurrence.	Plain safety logic; recurring-pain rule kept.
A3	Recovery after races / maximal efforts	Detected from RPE ≥ 9 , average HR ≥ 92 % of HRmax or pace ≥ 12 % faster than the runner's usual long runs, over ≥ 10 km (or a race in the calendar). Then no quality and no long run: ~7 days after a half marathon (first 2–3 days rest / cross-training), ~14 after a marathon, pro-rata below.	Convention in endurance practice; overload must be followed by recovery (Meeusen 2013). Exact days are a convention.
A4	Risk label consistent with the state	Přetížení or Tichý drift → at least „sledovat“; Kritická → „vysoké riziko“.	The state and the label must not contradict.
A5	Prior injury with unknown date = recent	Counted as ≤ 12 months until the runner fills in date and side (the profile asks for them); margins shrink as for a recent injury.	Previous injury is the most consistent risk factor (Hulme 2017).

Phase B — capacity that only grows with what was tolerated

#	Change	Rule	Basis
B1	Tolerated means confirmed	A session $> 1.3\times$ its capacity does not raise capacity for 14 days; afterwards only if a pain-free check-in or rating confirms it. Sessions with no report at all count at half weight („unknown“), not as tolerated.	Capacity = demonstrated tolerance (Bertelsen 2017); absence of a report is not evidence.
B2	Pain monitoring	Running with pain $\leq 5/10$ only if it settles by the next morning and does not rise week to week at the same site; higher next-morning pain or a rising trend → that load stops (flat, short, easy or no run).	Pain-monitoring model, RCT in Achilles tendinopathy (Silbernagel 2007); applied to other sites pragmatically.
B3	Injury record from the app	When A1/A2/B2 fire: „Je to zranění?“ → an injury record (date, site, side) that feeds injury history (margins) and a graded return to running.	OSTRC injury definition (Clarsen 2020).
B4	Race calendar	Several planned races with distance and priority (A/B/C). Warn on two race efforts within 14 days, and on racing all-out with readiness < 50 % or pain in the last 7 days; taper for A races.	Single-session spikes carry the risk (Frandsen 2025); taper (Bosquet 2007).

Phase C — credibility in the first weeks and a permanent check

#	Change	Rule	Basis
C1	First 6 weeks	No hard blocks from capacity channels that are not reliable yet; general rules instead: weekly volume $+10\text{--}15$ % at most, long run ≤ 30 -day longest $+ 10$ %; the 4-week cycle is placed only after 6 weeks.	Nielsen 2014; RUNSAFE (Frandsen 2025).
C2	Heart-rate zones	Birth year / measured HRmax in the profile; zones marked „odhad“ until then.	Z4+ minutes are only as good as HRmax.
C3	Validation replay as a tool	For every recorded injury: what the engine said 1–21 days before; run on the owner's data before each engine change, before vs after.	Guards against regressions; honest about $n = 1$.

4. Decisions taken

- Order: this document → phase A → phase B → phase C → open app feedback → one deploy.
- Check-in gets the three yes/no questions (A1). Recovery after races ~7 days (half marathon) / ~14 (marathon) (A3). A calendar of several races with priorities (B4). Unknown prior-injury date counts as recent (A5).
- After every change: backend, database (migrations for new columns / tables), API schemas and frontend checked together; full tests; the validation replay on the owner's data; screens checked on desktop and phone.

5. Expected result on the two events

- 25 Apr: „Přetížení · sledovat“ (not „nízké“), no long run; with the injury recorded, return-to-run rules afterwards instead of a hilly 16 km five days later.
- 12 Sep: the fast 21 km is a maximal effort — recovery block and no capacity gain until it is confirmed tolerated.
- 19 Sep: race warning (second race effort within 8 days, pain the week before); not a green light.
- 20–22 Sep: no running (acute overload after the race, pain at 3 sites, soreness 8, RPE 10), then easy only.
- Now: pain that limits movement → no running, physio recommended, injury record, graded return.

6. Implementation (engine v0.8.0) — refinements against the plan

Everything in sections 3–4 was built. Where the build had to decide a detail the plan left open, the decision is below.

#	As built	Why this way
B1	Only a JUMP (a session > 1.3× its own capacity that day) is held: it doesn't count for 14 days, then fully if a pain-free check-in or rating followed within 72 h, else at half weight. Sessions within 1.3× count as before. The weekly „best week“ skips windows > 1.3× the average week before them (judged only with ≥ 2 weeks of history before). The capacity panel shows a held jump and asks for a rating.	Half weight for every unreported session would have collapsed capacity for a runner who rarely rates runs (the owner has 4 check-ins and 6 ratings) — the risk sits in the jumps, not in ordinary runs.
B2	Morning worse: today's check-in ≥ 2/10 at a site vs yesterday's run rating there (a site the rating didn't mark counts as 0) → no running today. Trend: mean of each day's highest pain, last 7 days vs the 7 before, +1 and ≥ 2 → no intensity / long run; needs ≥ 2 report days now and ≥ 1 before.	1/10 is noise; days without reports are unknown, not pain-free.
B3	A report with OSTRC severity > 0 (self or physio-confirmed) adds its site to the prior injury; a new episode sets its date and side (the same episode keeps the start). After „zahojení“: 3 weeks at 50 / 75 / 90 % of the average week of the 4 weeks before the injury, no intensity for 14 days. „Je to zranění?“ opens the OSTRC form prefilled with the painful sites.	Injury history no longer depends on the profile being filled in by hand.
B4	Race calendar (date, name, km, priority A/B/C); the profile's goal race counts as an A race. Taper and the pre-race week follow the next A race only; race day applies to any race, but an override (injury, limited function, pain worse overnight, acute overload) still wins. Warnings: an A/B race ≤ 14 days after a maximal effort or another A/B race; on race day / the eve readiness < 50 % or pain ≥ 2 in the last 7 days.	C races are training; a warning on the eve leaves time to decide.
C1	First 42 days of data: last week + 10 %, long run ≤ 1.1 × the longest of 30 days, 10 min of hard work in a quality session, intensity / climbing / total-load channels don't limit, no (manual) cycle.	As planned.
C2	Measured HR max in the profile replaces the estimate, never below the hardest average HR recorded; zones say „odhad“ until then.	On the owner's data the estimate (≈ 185) put 86 min/week in Z5; with 193 it is 21.
C3	app/metrics/validation.py — injury_timeline() / python -m app.metrics.validation [date]. Morning view; a prior injury is hidden before its own date.	Otherwise the April injury would „predict itself“.
A4+	No quality session while the state is Přetížení / Kritická, also inside the hysteresis band (load < 25).	19 Sep: the state said Přetížení, the plan still offered a quality session.

Also in this release (in-app feedback): sleep quality in readiness — deep + REM share and efficiency against the runner's 8-week normal, at most half a signal because watch staging is approximate (stages are backfilled once from Garmin); a run can be excluded

from every calculation and restored; the Zátěž tab shows the weekly safe longest run (not scaled by today's readiness).

7. Replay after the changes (owner's data, v0.8.0)

Event	First warning	Days warned before	On the day	Remaining gap
ITBS, 25 Apr	20 Apr (5 days)	9 of 21	Přetížení · sledovat, volno (rest)	The run happened against a rest recommendation — the app can only advise.
Groin / adductor, 19 Sep	31 Aug (19 days)	19 of 21	Přetížení · sledovat, lehký běh (was kvalitní)	12 Sep: volno recommended; with the race in the calendar the taper starts 5 Sep and the warning „race 7 days after a maximal effort“ shows from 12 Sep. Pain before the race was never logged, so B2 couldn't fire.

What still depends on the runner: logging pain and the three function questions in the check-in, rating runs (confirms jumps as capacity), entering races, a measured HR max, and the injury date and side in the profile.

References

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